

Crown Change Equations for CONUS

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Data

We took the FIA crown change data subset for all species, filtered out missing data and trees with negative height to live crown change (Δh_{tlc}). This left 2215065 observations.

The observations are distributed among states as shown in Table 1.

Table 1: Height to Live Crown Change Observations by State

State	n
MN	164734
WI	155631
OR	139365
MI	134501
ME	129550
GA	116808
WA	89806
AL	87674
MS	73224
VA	72765
NC	72008
NY	69882
SC	68245
CA	67734
PA	65135
AR	62112
MO	60823
TN	59729
TX	50805
MT	49059
KY	45873

Table 1: Height to Live Crown Change Observations by State

State	n
WV	35546
NH	33683
ID	33557
FL	29884
CO	29436
LA	27835
VT	24785
OH	24151
OK	21188
IN	19062
IL	13474
UT	11570
MA	11190
IA	8157
AZ	7907
NJ	6279
MD	5933
SD	5851
KS	5660
NM	4855
CT	4728
WY	4465
NE	2893
ND	2250
RI	2240
DE	2012
NV	1011

The observations are distributed among species as shown in Table 2.

Table 2: Height to Live Crown Change Observations by Species

SPCD	CommonName	n
131	loblolly pine	241676
316	red maple	147268
202	Douglas-fir	116548
318	sugar maple	81946
746	quaking aspen	79948

Table 2: Height to Live Crown Change Observations by Species

SPCD	CommonName	n
12	balsam fir	67214
611	sweetgum	56431
802	white oak	55650
108	lodgepole pine	51560
122	ponderosa pine	50778
241	northern white-cedar	48237
129	eastern white pine	40097
621	yellow-poplar	40027
833	northern red oak	37327
125	red pine	35601
111	slash pine	34291
261	eastern hemlock	33717
531	American beech	30834
762	black cherry	28830
95	black spruce	27283
110	shortleaf pine	27185
97	red spruce	27097
832	chestnut oak	26981
263	western hemlock	25716
19	subalpine fir	25489
375	paper birch	25037
835	post oak	23248
837	black oak	22698
68	eastern redcedar	22236
827	water oak	22124
371	yellow birch	20935
693	blackgum	20633
543	black ash	20068
541	white ash	19547
15	white fir	19412
93	Engelmann spruce	17820
17	grand fir	17703
951	American basswood	16967
972	American elm	15973
743	bigtooth aspen	15894
544	green ash	15766
71	tamarack (native)	15615
403	pignut hickory	14513
409	mockernut hickory	14245

Table 2: Height to Live Crown Change Observations by Species

SPCD	CommonName	n
711	sourwood	13821
121	longleaf pine	13451
11	Pacific sliver fir	13270
132	Virginia pine	13070
372	sweet birch	13021
94	white spruce	12027
971	winged elm	11955
812	southern red oak	11855
806	scarlet oak	11686
105	jack pine	11529
242	western redcedar	10043
407	shagbark hickory	9837
823	bur oak	9188
701	eastern hophornbeam	8852
631	tanoak	8177
264	mountain hemlock	6833
602	black walnut	6738
73	western larch	6306
81	incense-cedar	6249
694	swamp tupelo	6219
931	sassafras	6035
402	bitternut hickory	5938
820	laurel oak	5856
741	balsam poplar	5549
591	American holly	5429
408	black hickory	5380
491	flowering dogwood	5239
462	hackberry	5195

Proposed Model

The model we propose to estimate $\Delta htlc$ is Equation 1.

$$\Delta htlc = cl(1 - e^{(\beta_0 + \beta_1 \Delta ht + \beta_2 cch)}) \quad (1)$$

where:

- $\Delta htlc$: change in height to live crown (feet)

- Δht : height growth (feet)
- cl : crown length (feet)
- cch : crown closure at tree tip (fraction of acre)

The rationale for the model form is:

- Change in height to live crown is bound by cl which is the maximum change in $htlc$ that is possible (the tree loses its entire crown).
- Change is driven by current height growth (Δht) which was found to be strongly correlated and likely to be associated with crown dynamics (both building and receding crown) and by cch which is a measure of the competition the tree sees in its peers and superiors.

Our expectation is as Δht and cch increase, $\Delta htlc$ will increase.

We used an integrative fitting approach to account for varying period lengths.

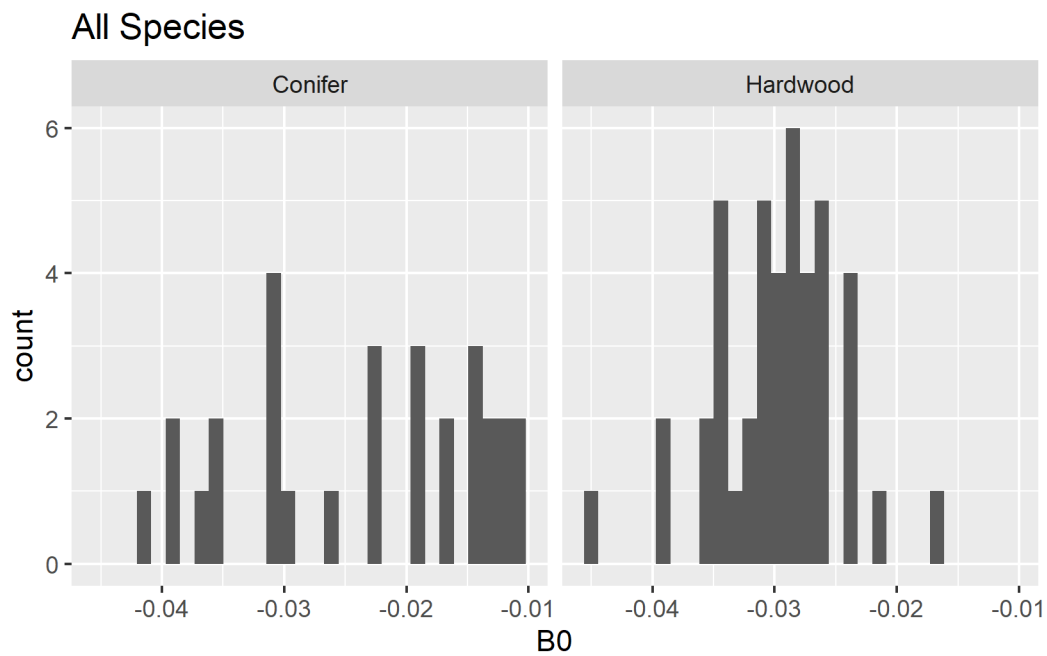
Table 3: Parameter Estimates

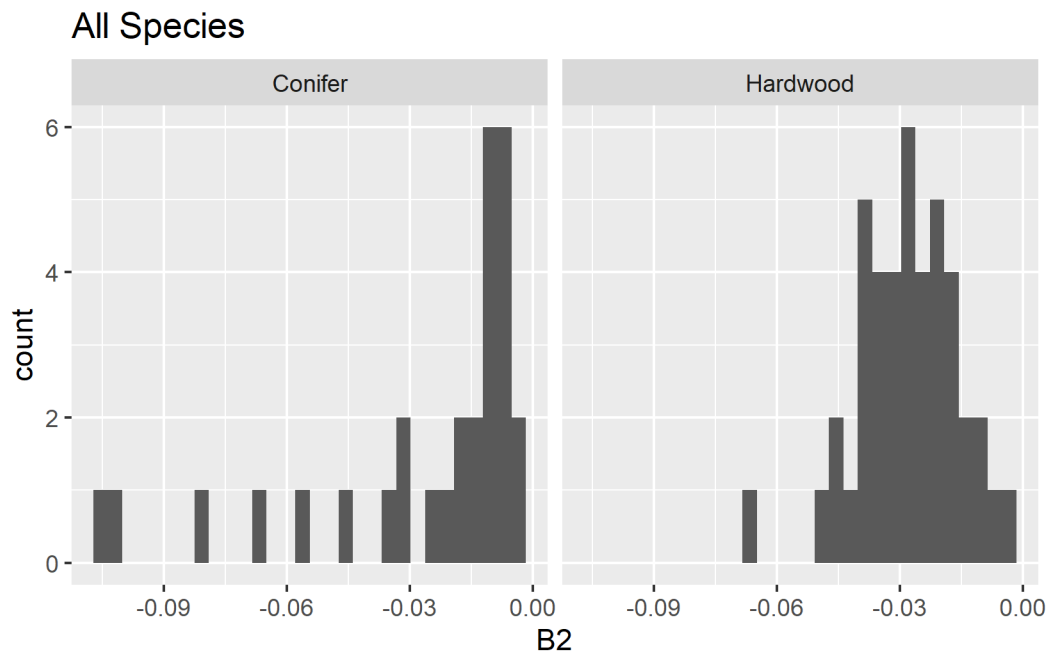
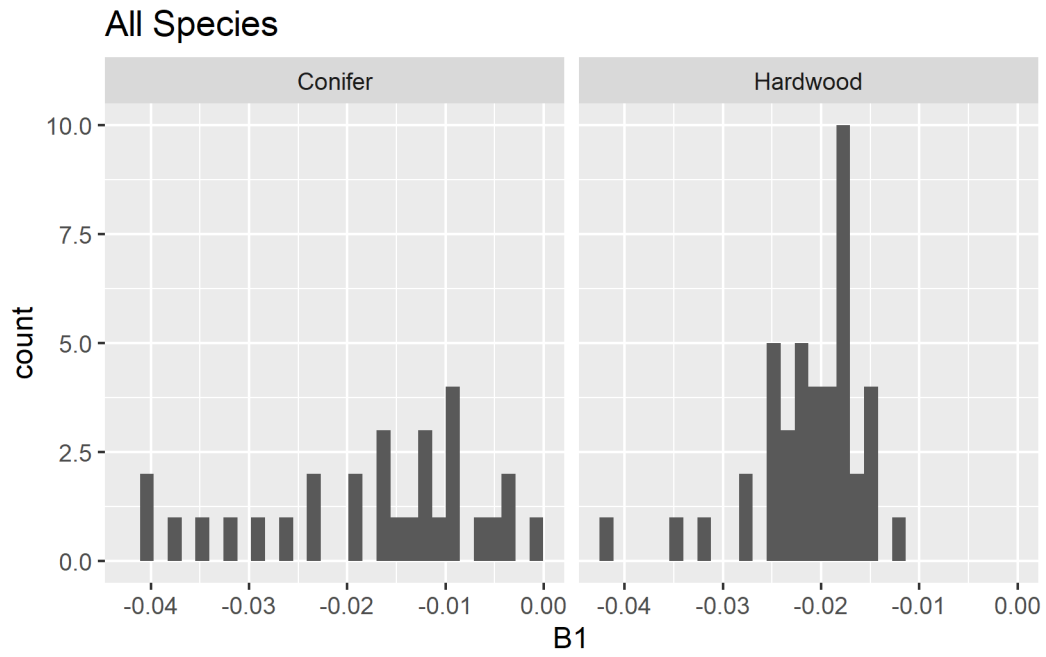
SPCD	CommonName	n	b0	b1	b2	rmse
131	loblolly pine	241676	-0.0132069	-0.0408399	-0.1053717	4.863517
316	red maple	147268	-0.0258713	-0.0193705	-0.0240361	4.346895
202	Douglas-fir	116548	-0.0112234	-0.0156854	-0.0255942	8.501589
318	sugar maple	81946	-0.0241431	-0.0160077	-0.0184761	4.583287
746	quaking aspen	79948	-0.0314025	-0.0278855	-0.0259455	4.024670
12	balsam fir	67214	-0.0418278	-0.0099556	-0.0045387	4.235663
611	sweetgum	56431	-0.0334204	-0.0208898	-0.0281521	5.026648
802	white oak	55650	-0.0275070	-0.0154478	-0.0335570	4.912433
108	lodgepole pine	51560	-0.0170380	-0.0152015	-0.0127866	4.594667
122	ponderosa pine	50778	-0.0131708	-0.0098795	-0.0200531	5.712518
241	northern white-cedar	48237	-0.0292742	-0.0236157	-0.0115457	3.734431
129	eastern white pine	40097	-0.0307521	-0.0162628	-0.0320762	4.902369
621	yellow-poplar	40027	-0.0298053	-0.0172139	-0.0408220	5.676039
833	northern red oak	37327	-0.0240785	-0.0174521	-0.0384571	4.602915
125	red pine	35601	-0.0312523	-0.0288393	-0.0183529	3.930194
111	slash pine	34291	-0.0220847	-0.0397923	-0.1021818	4.416290
261	eastern hemlock	33717	-0.0262842	-0.0119701	-0.0082307	4.791652
531	American beech	30834	-0.0343865	-0.0121317	-0.0048302	5.266936
762	black cherry	28830	-0.0313557	-0.0270923	-0.0395168	4.442232
95	black spruce	27283	-0.0395152	-0.0190016	-0.0086078	4.191788
110	shortleaf pine	27185	-0.0227298	-0.0320556	-0.0651290	3.961519
97	red spruce	27097	-0.0302922	-0.0184631	-0.0147330	3.428595
832	chestnut oak	26981	-0.0275195	-0.0174546	-0.0367974	4.530335
263	western hemlock	25716	-0.0125214	-0.0123315	-0.0075014	8.038939

SPCD	CommonName	n	b0	b1	b2	rmse
19	subalpine fir	25489	-0.0190843	-0.0007513	-0.0055258	5.612792
375	paper birch	25037	-0.0309199	-0.0237790	-0.0258117	4.014620
835	post oak	23248	-0.0284455	-0.0204090	-0.0297372	3.716655
837	black oak	22698	-0.0287366	-0.0182102	-0.0393694	4.631588
68	eastern redcedar	22236	-0.0304116	-0.0232003	-0.0190284	3.957937
827	water oak	22124	-0.0320482	-0.0191583	-0.0338594	5.137682
371	yellow birch	20935	-0.0235464	-0.0183461	-0.0189050	4.044320
693	blackgum	20633	-0.0321264	-0.0224974	-0.0208703	4.161153
543	black ash	20068	-0.0385451	-0.0245387	-0.0230414	4.153443
541	white ash	19547	-0.0338075	-0.0207186	-0.0318036	4.887246
15	white fir	19412	-0.0170875	-0.0048399	-0.0099576	7.184766
93	Engelmann spruce	17820	-0.0142027	-0.0039424	-0.0075208	5.691465
17	grand fir	17703	-0.0193986	-0.0041256	-0.0113590	7.466754
951	American basswood	16967	-0.0293754	-0.0194890	-0.0220865	4.512422
972	American elm	15973	-0.0351687	-0.0230249	-0.0151259	4.240959
743	bigtooth aspen	15894	-0.0309933	-0.0244155	-0.0665545	4.558139
544	green ash	15766	-0.0343653	-0.0225335	-0.0314910	4.733937
71	tamarack (native)	15615	-0.0359264	-0.0168823	-0.0322782	4.708876
403	pignut hickory	14513	-0.0288621	-0.0149961	-0.0286518	5.024891
409	mockernut hickory	14245	-0.0286647	-0.0155044	-0.0278595	4.626932
711	sourwood	13821	-0.0280695	-0.0320836	-0.0179879	3.622674
121	longleaf pine	13451	-0.0230458	-0.0371968	-0.0807762	4.424798
11	Pacific sliver fir	13270	-0.0148760	-0.0091696	-0.0067639	6.150729
132	Virginia pine	13070	-0.0355920	-0.0343352	-0.0549796	4.488140
372	sweet birch	13021	-0.0208677	-0.0242859	-0.0191999	4.422035
94	white spruce	12027	-0.0364690	-0.0138824	-0.0116295	4.956808
971	winged elm	11955	-0.0346135	-0.0220443	-0.0171053	3.934510
812	southern red oak	11855	-0.0277253	-0.0190194	-0.0333075	4.897130
806	scarlet oak	11686	-0.0262224	-0.0174783	-0.0454733	4.785137
105	jack pine	11529	-0.0388025	-0.0261864	-0.0466200	3.778647
242	western redcedar	10043	-0.0108020	-0.0069198	-0.0090418	6.785922
407	shagbark hickory	9837	-0.0262590	-0.0165618	-0.0265234	4.710684
823	bur oak	9188	-0.0351170	-0.0150922	-0.0281166	4.334088
701	eastern hophornbeam	8852	-0.0289326	-0.0246251	-0.0118204	3.717221
631	tanoak	8177	-0.0169150	-0.0217326	-0.0087146	5.033697
264	mountain hemlock	6833	-0.0123877	-0.0089012	-0.0033705	5.768624
602	black walnut	6738	-0.0259921	-0.0216733	-0.0467813	4.592716
73	western larch	6306	-0.0190439	-0.0116612	-0.0342672	6.881039
81	incense-cedar	6249	-0.0145741	-0.0095402	-0.0115009	5.960956
694	swamp tupelo	6219	-0.0314106	-0.0228373	-0.0306356	4.317265
931	sassafras	6035	-0.0296592	-0.0344279	-0.0366042	3.969130

SPCD	CommonName	n	b0	b1	b2	rmse
402	bitternut hickory	5938	-0.0269162	-0.0173432	-0.0378208	4.925288
820	laurel oak	5856	-0.0338921	-0.0174190	-0.0327430	5.287426
741	balsam poplar	5549	-0.0450147	-0.0247054	-0.0477673	4.079945
591	American holly	5429	-0.0258857	-0.0179057	-0.0081779	3.572879
408	black hickory	5380	-0.0292102	-0.0182325	-0.0194525	3.930248
491	flowering dogwood	5239	-0.0387001	-0.0418162	-0.0122040	2.653423
462	hackberry	5195	-0.0237831	-0.0204403	-0.0216425	3.913652

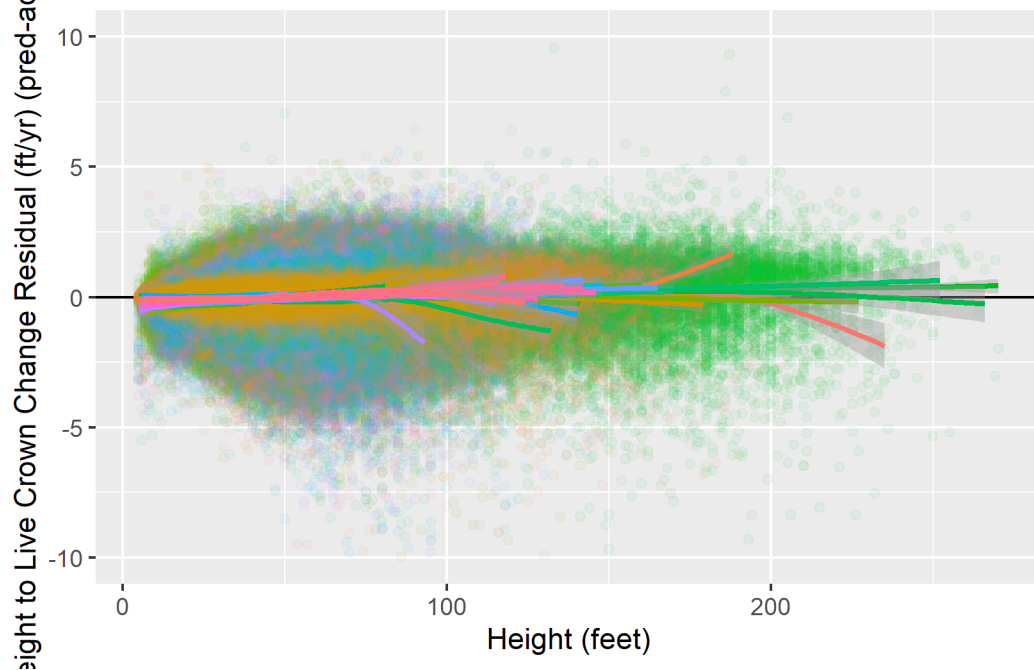
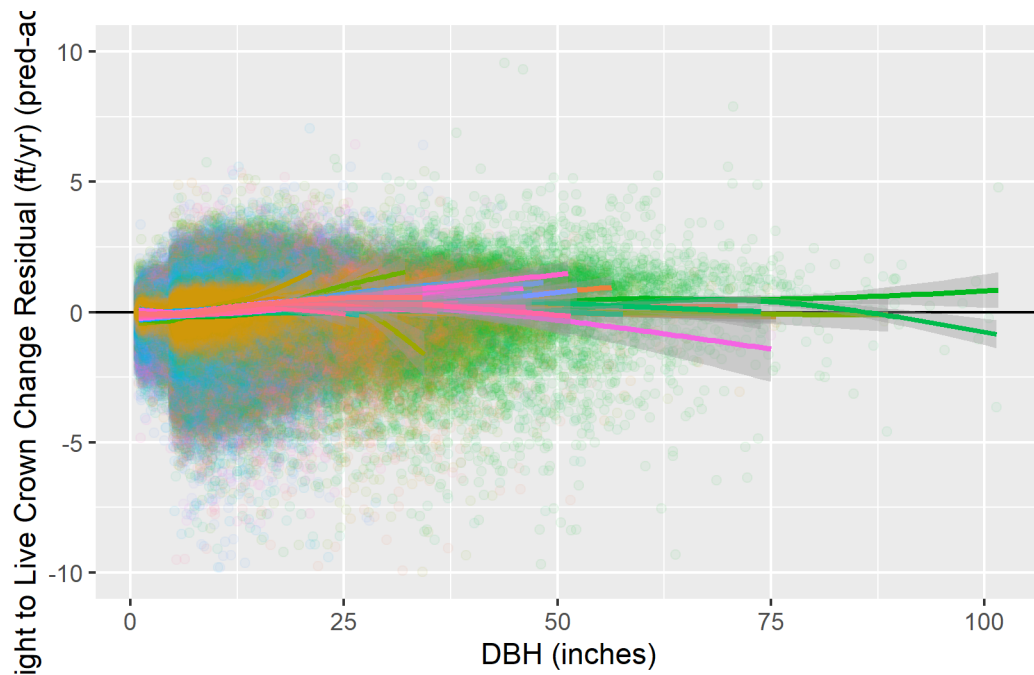
We were able to make base rate estimates for 72 species. The distributions of the estimates for these species are shown below.

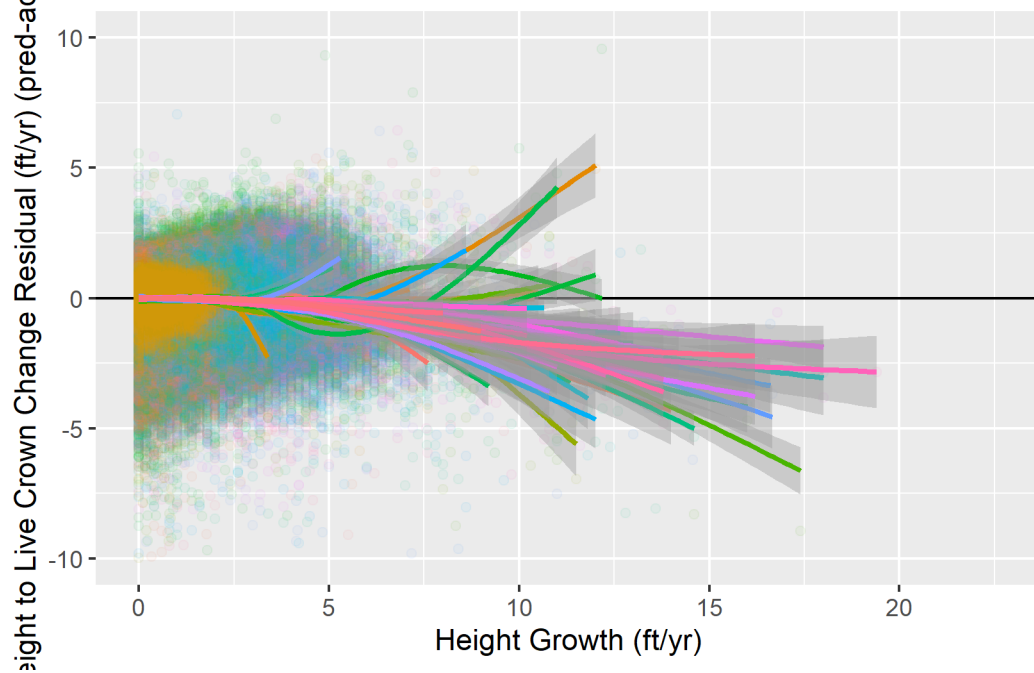
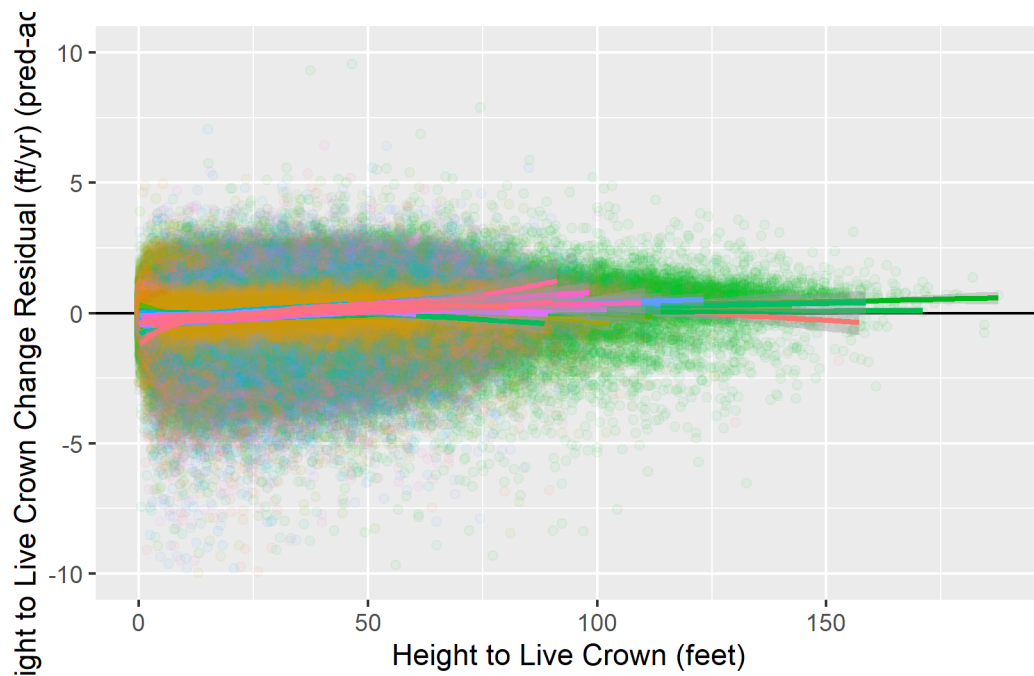


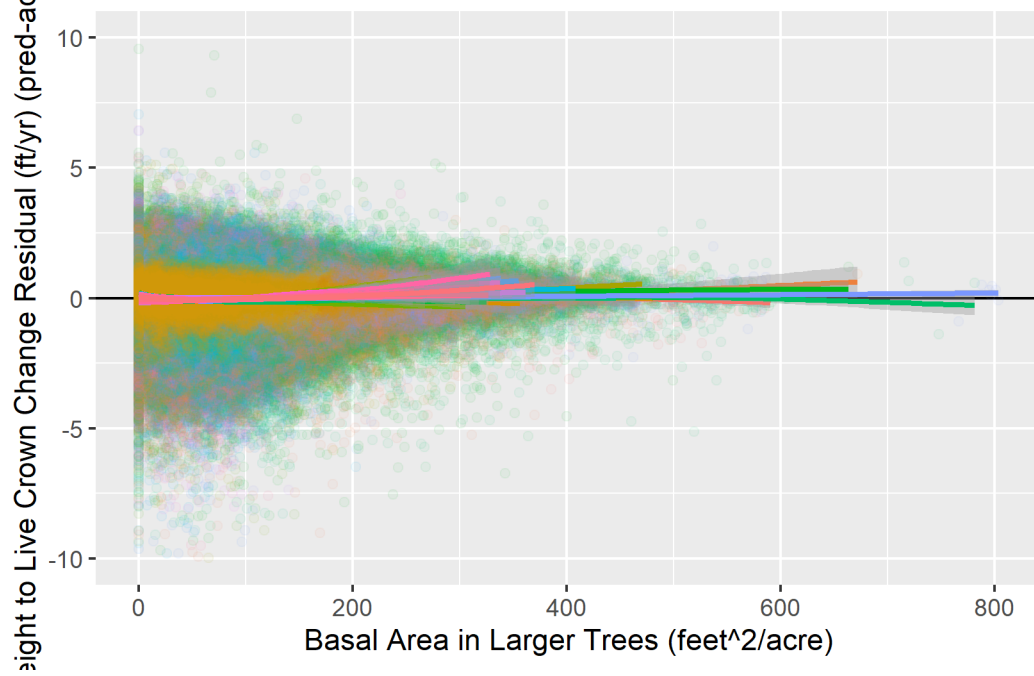
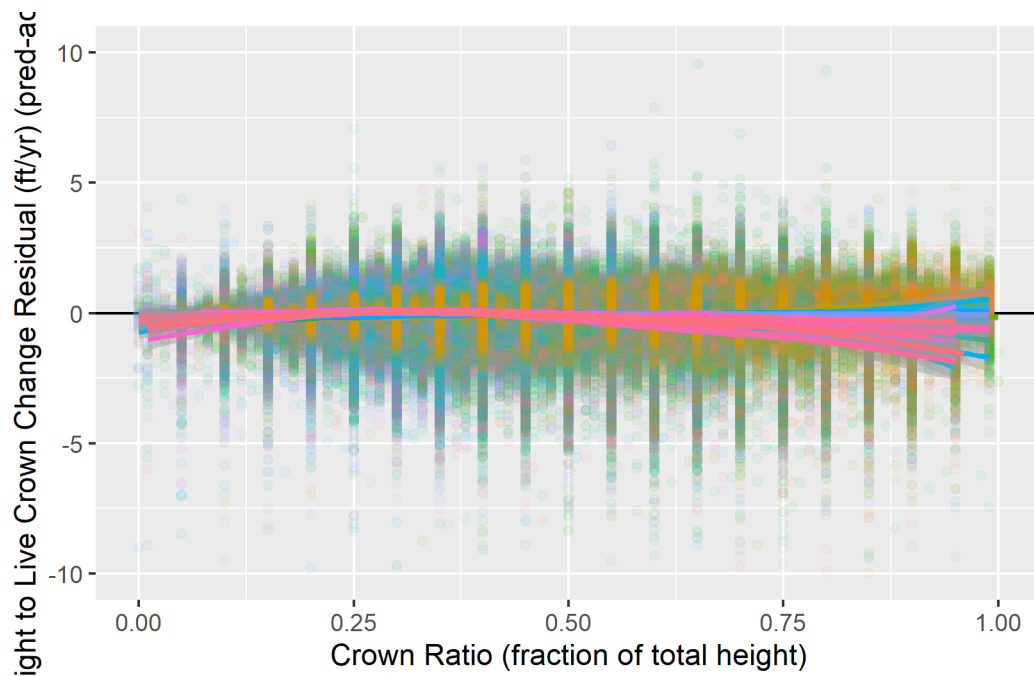


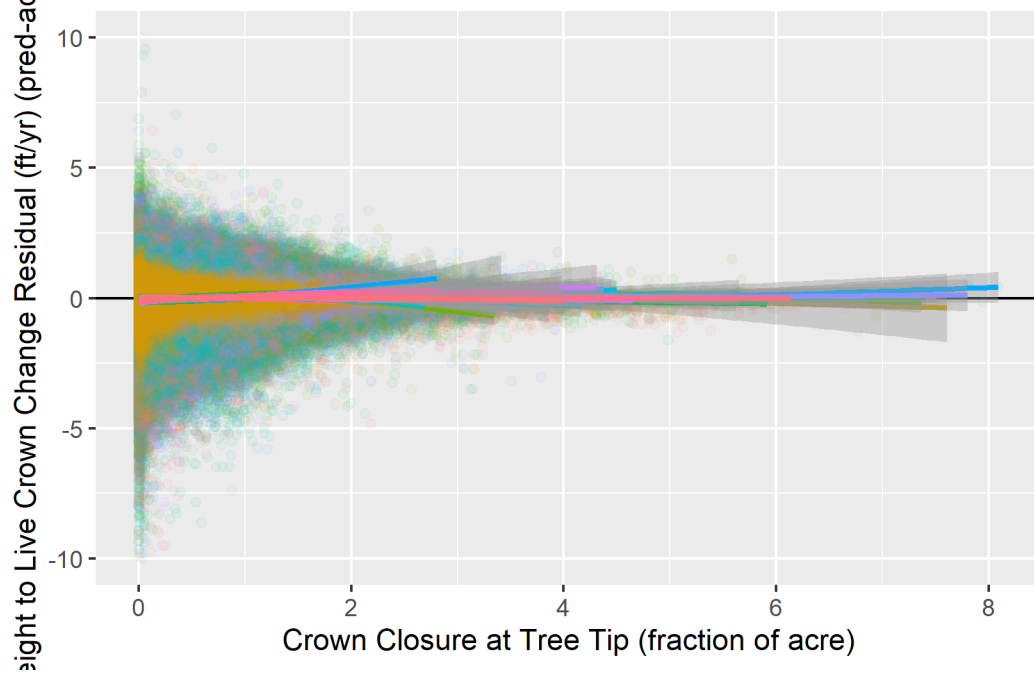
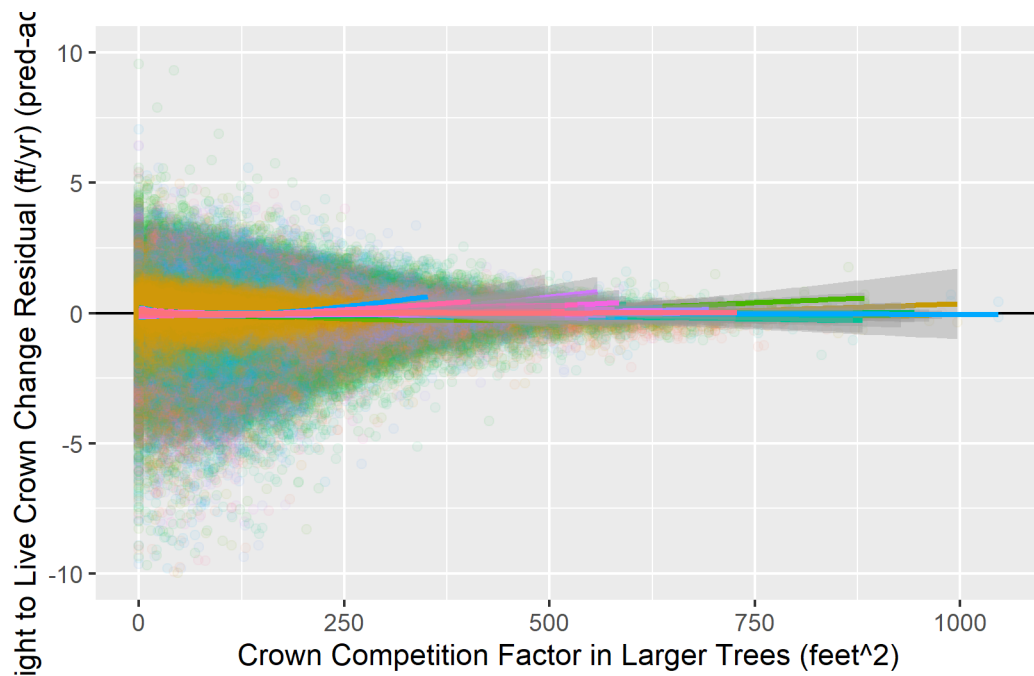
Height to Live Crown Residuals

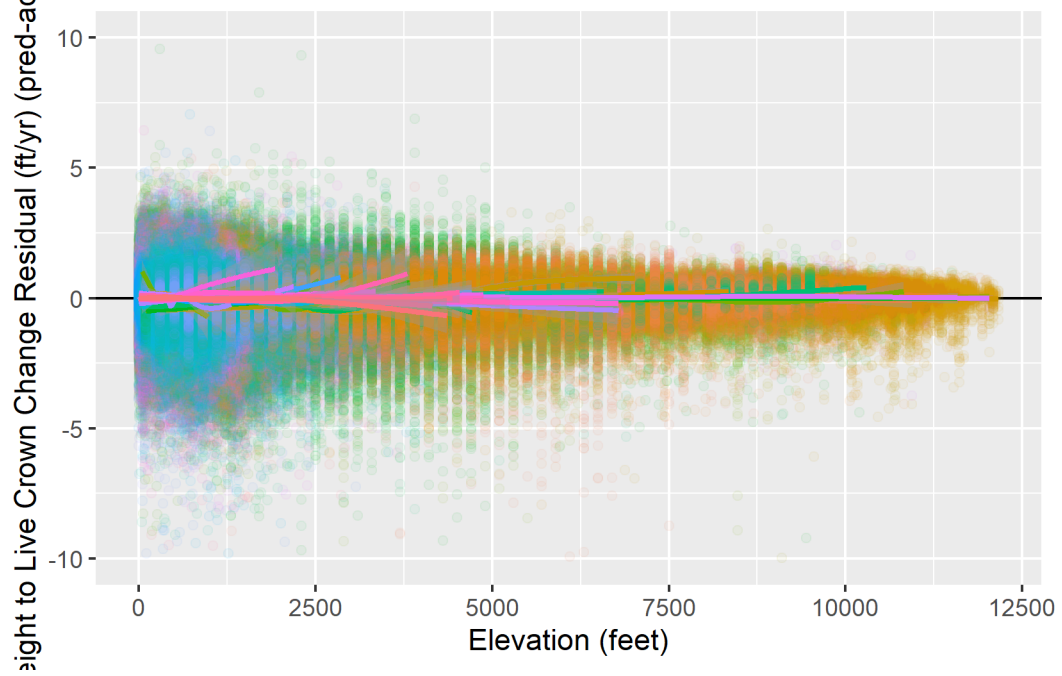
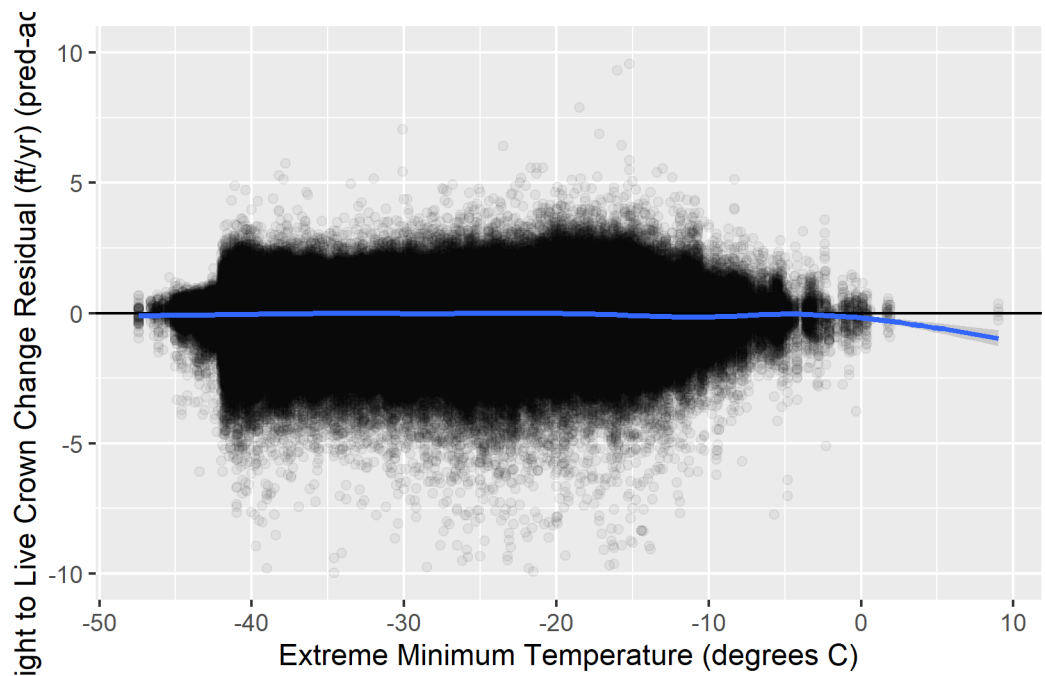
The following residual graphs show the errors for all fitted species using the proposed model.

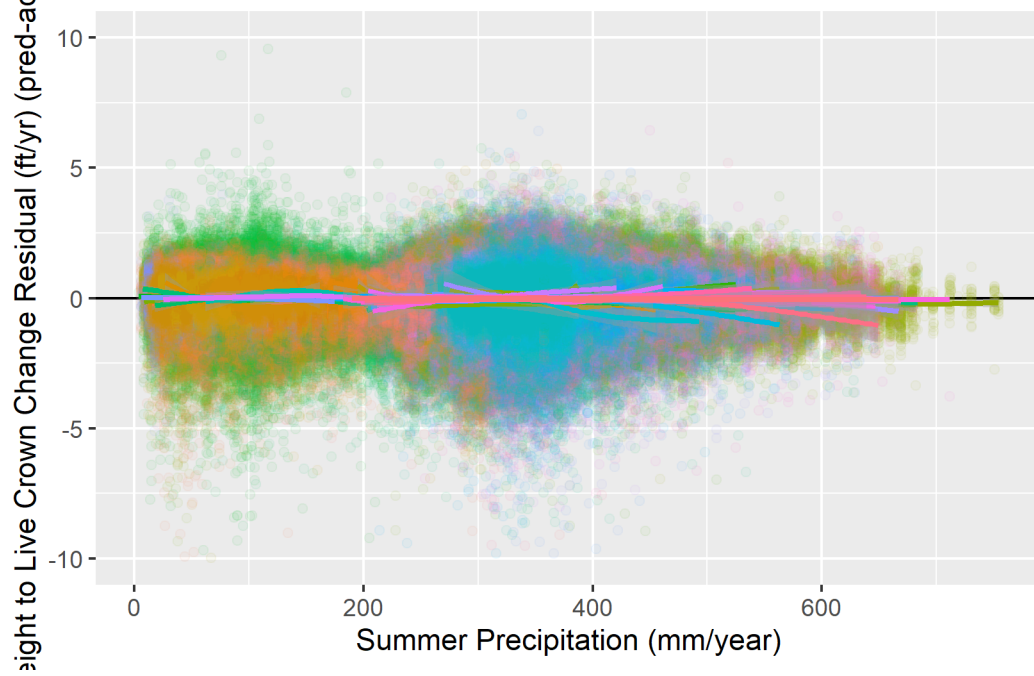
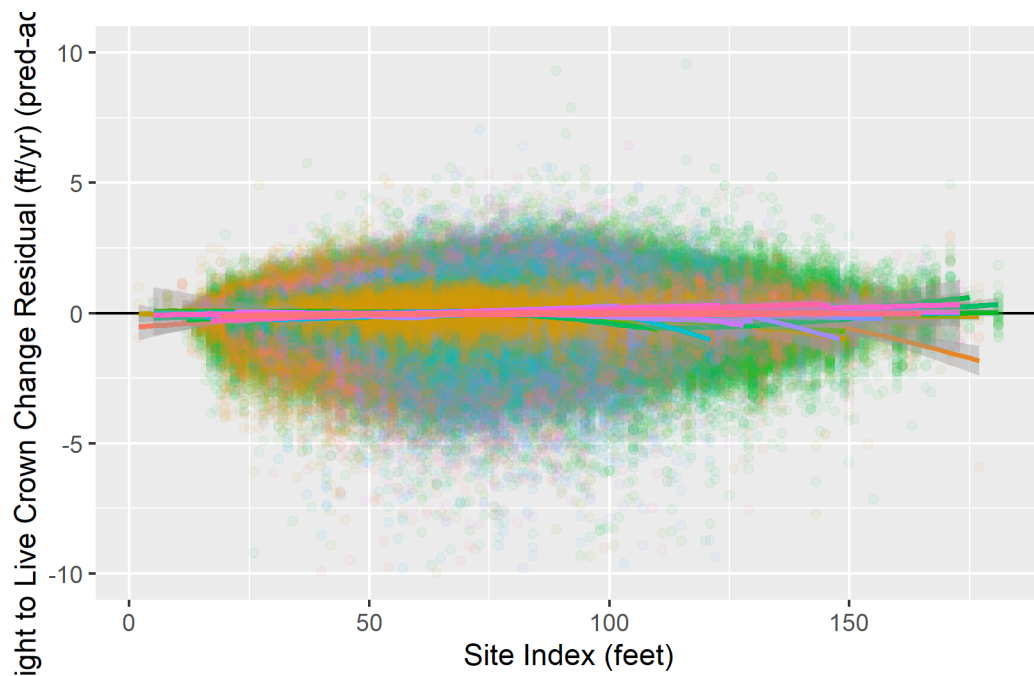


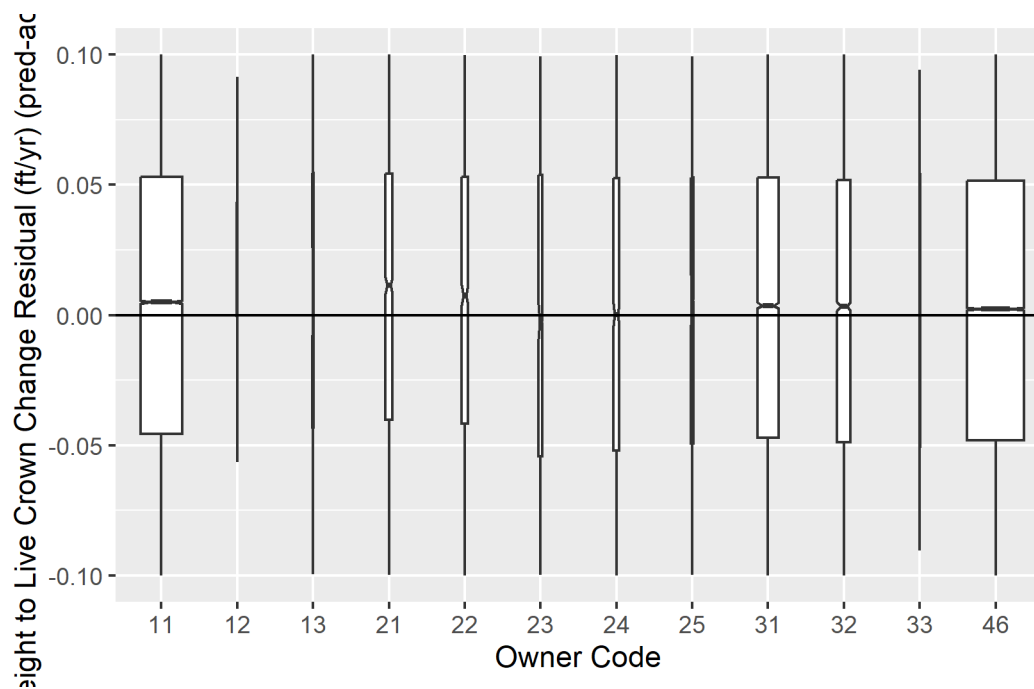












State - Unit - County Mean Height to Live Crown Change Residual

